

23. Optical Properties of Metals and Inelastic Scattering

23.1 Introduction

The interaction of electromagnetic radiation with metals divides between low and high frequencies. Low frequency refers to frequencies too small to induce transitions from one Bloch band to another. The energies involved in hopping between bands are characteristically around 1 eV, so the dividing line is around

$$1 \text{ eV} \Rightarrow \omega \sim 10^{15} \text{ Hz} \Rightarrow \lambda \sim 1 \text{ } \mu\text{m}. \quad (23.1)$$

For frequencies below this point, the absence of transitions between bands means that the behavior of electrons is semiclassical and can be treated by the formalism of Section 17.2. This is not to say that there are no quantum-mechanical effects; the response of electrons even to static electric fields has many quantum-mechanical features, but classical notions of electrons wiggling back and forth in response to oscillating fields provide the right starting point. At frequencies above the cutoff (23.1), the language and mode of thought in discussing the problem begin to alter. Although the incoming light will still be treated classically, and there is no need to quantize the electromagnetic field, the effects of the light waves are most easily understood as a beam of photons whose absorption is able to induce transitions between various quantum mechanical states. For energies of 10–100 eV, light has become the most powerful probe of electronic structure in metals and semiconductors, making possible in some cases a direct experimental investigation of the one-electron picture, including quantitative measurements of the energy bands.

23.1.1 Plasma Frequency

In the Drude model, a metal consists of a gas of noninteracting mobile electrons. The response of such electrons to external electric fields was described in Eqs. (20.4) and (20.31), which showed that

$$\epsilon(\omega) = 1 - \frac{\omega_p^2}{\omega(\omega + i/\tau)} \quad \begin{array}{l} \text{To compare with experiment, the constant 1} \\ \text{often must be replaced by } \epsilon^\infty. \end{array} \quad (23.2)$$

$$\omega_p = \sqrt{\frac{4\pi ne^2}{m}} = 5.64 \cdot 10^{15} \text{ Hz} \left[\frac{n}{10^{22} \text{ cm}^{-3}} \right]^{1/2}. \quad (23.3)$$

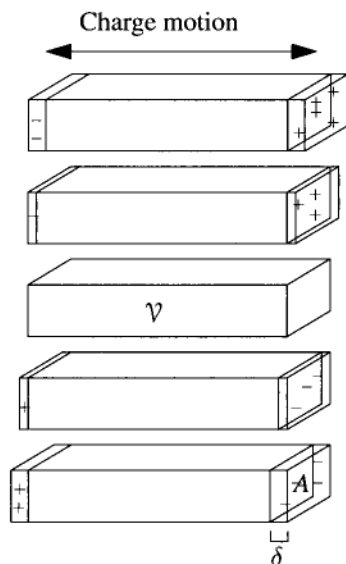


Figure 23.1. Plasma oscillations result from the bulk motion of charge in a solid.

The definition of the plasma frequency ω_p is motivated by the limit $\omega\tau \gg 1$, which leads to

$$\epsilon(\omega) = 1 - \left(\frac{\omega_p}{\omega} \right)^2. \quad (23.4)$$

It has a very simple physical interpretation, illustrated in Figure 23.1. Consider a slab of electrons sitting on top a slab of neutralizing positive charge. If the whole slab of electrons is moved a distance δ , then positive charge develops on one side of the slab, negative charge on the other. The charge, say, on the positive side is $enA\delta$, where A is the area of the slab, and n the density of charged particles. A uniform electric field $E = 4\pi en\delta$ develops between two such slabs of charge. If the total volume of the system is $\mathcal{V}$, then the force on all of the electrons is

$$en\mathcal{V}E = 4\pi n^2 e^2 \mathcal{V} \delta. \quad (23.5)$$

Setting this force equal to mass, $mn\mathcal{V}$, times acceleration $\ddot{\delta}$ gives

$$\ddot{\delta} = - \frac{4\pi ne^2}{m} \delta. \quad (23.6)$$

So the frequency of plasma oscillations is

$$\omega_p^2 = \frac{4\pi ne^2}{m}. \quad (23.7)$$

This result is correct only for free electrons. In the presence of a periodic potential, the plasma frequency is modified. Conventionally, the electron mass m is replaced by an effective mass called the optical mass; see Eq. (23.27).

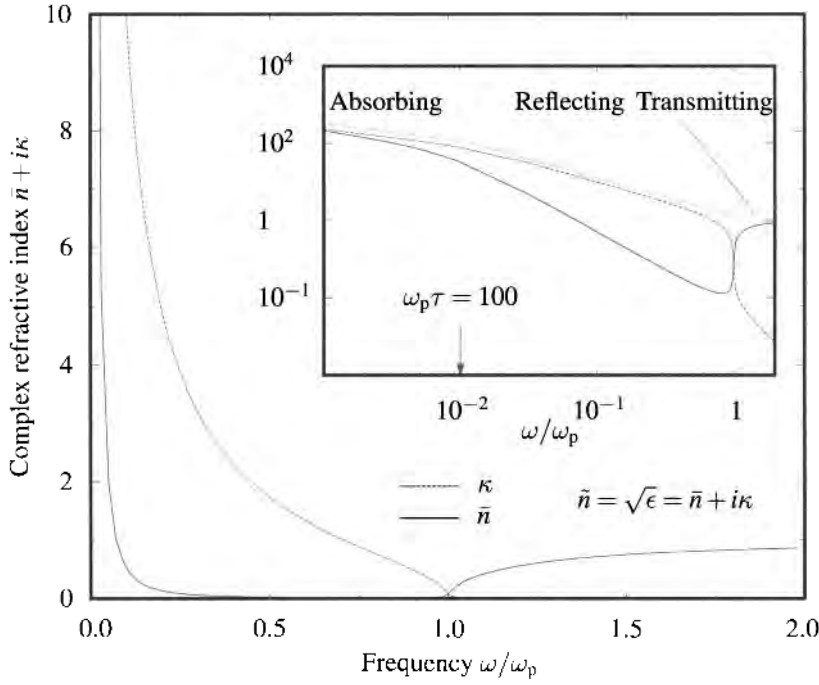


Figure 23.2. Index of refraction $\bar{n}$ and extinction coefficient κ for metal obeying Eq. (23.2), with $\omega_p\tau = 100$, showing absorbing, reflecting, and transmitting frequency ranges.

Three Regimes for Metallic Response. The product $\omega_p\tau$ can be estimated from Eqs. (16.7) and (23.3) and can range from 10 to 100. Therefore, according to Eq. (23.2), the optical absorption of metals should pass through three phases, shown in Figure 23.2.

Absorbing: For $0 < \omega\tau \lesssim 1$ the metal absorbs the incident radiation, because

$$\epsilon \approx 1 + i\tau \frac{\omega_p^2}{\omega} (1 + i\omega\tau) \Rightarrow \bar{n} \approx \kappa \approx \sqrt{\tau\omega_p^2/2\omega}. \quad (23.8)$$

The index of refraction $\bar{n}$ and extinction coefficient κ were defined in Eqs. (20.18) and (20.20).

Reflecting: For $1 \lesssim \omega\tau \lesssim \omega_p\tau$ the real part of ϵ is negative, equaling approximately

$$\frac{(\omega^2 - \omega_p^2)}{\omega^2} \quad (23.9)$$

and resulting in

$$\kappa \approx \sqrt{\omega_p^2/\omega^2 - 1} \quad \text{and} \quad \bar{n} \approx \frac{\omega_p^2}{2\tau\omega^2 \sqrt{\omega_p^2 - \omega^2}}. \quad (23.10)$$

According to Eq. (20.43), because the index of refraction $\bar{n}$ is small in comparison with the extinction coefficient κ , almost all radiation incident upon the metal in this range of frequencies will be reflected.

Transparent: For $\omega \gtrsim \omega_p$ the real part of the dielectric constant first vanishes and then becomes positive as ω increases. The metal becomes nearly transparent to the incoming radiation, with index of refraction and extinction coefficient given by

$$\bar{n} \approx \sqrt{1 - \omega_p^2/\omega^2} \quad \kappa \approx \frac{\omega_p^2}{2\tau\omega^2 \sqrt{\omega^2 - \omega_p^2}}. \quad (23.11)$$

23.2 Metals at Low Frequencies

When the frequency of radiation is low enough that the semiclassical approach is valid, a restriction that was the subject of Problem 20.2, the population of electrons is described by the Boltzmann equation (Section 17.2), which, in the absence of thermal gradients reads

$$\frac{\partial g}{\partial t} = -\vec{v} \cdot \nabla g - e\vec{E} \cdot \vec{v} \frac{\partial g}{\partial \mu} - \frac{g}{\tau}. \quad \text{Use } \partial f / \partial \vec{k} = \partial \mathcal{E} / \partial \vec{k} \partial f / \partial \mathcal{E} N = \dots \hbar \omega \partial f / \partial \mu, \quad (23.12)$$

starting with Eq. (17.10) in the relaxation time approximation.

Suppose the electric field to have the form

$$\vec{E} = \vec{E}(\vec{q}, \omega) e^{i\vec{q} \cdot \vec{r} - i\omega t}. \quad (23.13)$$

Then Eq. (23.12) is easily solved by taking

$$g_{\vec{k}} = g_{\vec{k}}(\vec{q}, \omega) e^{i\vec{q} \cdot \vec{r} - i\omega t} \quad \text{The label } \vec{k} \text{ describes the Bloch index of an electron, while the label } \vec{q} \text{ describes the wave vector of the electromagnetic wave.} \quad (23.14)$$

so that

$$g_{\vec{k}}(\vec{q}, \omega)[-i\omega] = [-i\vec{v} \cdot \vec{q} - 1/\tau] g_{\vec{k}}(\vec{q}, \omega) - e\vec{E} \cdot \vec{v} \frac{\partial f}{\partial \mu} \quad (23.15)$$

$$\Rightarrow g_{\vec{k}}(\vec{q}, \omega) = -e \frac{\partial f}{\partial \mu} \frac{\vec{E}(\vec{q}, \omega) \cdot \vec{v}}{1/\tau - i(\omega - \vec{q} \cdot \vec{v})}. \quad \text{Remember that } \vec{v} \text{ is a function of } \vec{k} \text{ according to Eq. (7.59).} \quad (23.16)$$

The current associated with the distribution function (23.16) is determined by an integral over $\vec{k}$, and from it can be obtained the conductivity σ through

$$\vec{j} = -e \int [d\vec{k}] \vec{v} g_{\vec{k}} \quad [d\vec{k}] \text{ defined in Eq. (6.15).} \quad (23.17)$$

$$= e^2 \int [d\vec{k}] \frac{\partial f}{\partial \mu} \frac{\vec{v} [\vec{v} \cdot \vec{E}(\vec{q}, \omega)]}{1/\tau - i(\omega - \vec{q} \cdot \vec{v})} \quad (23.18)$$

$$\Rightarrow \sigma_{\alpha\beta} = e^2 \int [d\vec{k}] \frac{\partial f}{\partial \mu} \frac{v_\alpha v_\beta}{1/\tau - i(\omega - \vec{q} \cdot \vec{v})} \quad \text{The subscript on } v \text{ is a Cartesian coordinate.} \quad (23.19)$$

$$= e^2 \int \frac{d\Sigma}{4\pi^3 \hbar v} \frac{v_\alpha v_\beta}{1/\tau - i(\omega - \vec{q} \cdot \vec{v})} \quad \text{The integral is over the Fermi surface, as in Eq. (7.73). Use Eq. (6.15).} \quad (23.20)$$

$$\Rightarrow \epsilon_{\alpha\beta} = \delta_{\alpha\beta} + \frac{4\pi e^2}{\omega} \int \frac{d\Sigma}{4\pi^3 \hbar v} \frac{v_\alpha v_\beta}{1/\tau - i(\omega - \vec{q} \cdot \vec{v})} \quad \text{Using Eq. (20.14) to relate conductivity to dielectric constant.} \quad (23.21)$$

The wave vector $\vec{q}$ of transverse propagating electromagnetic waves can now be determined from

$$q = \frac{\sqrt{\epsilon}\omega}{c} = (\bar{n} + i\kappa) \frac{\omega}{c}. \quad (23.22)$$

Because $\vec{q}$ appears in the middle of the integral (23.21), this task is not particularly simple.

Recovery of Drude Formula. Often, the $\vec{q}$ dependence of (23.21) can be neglected. The conditions needed to neglect it are that $\vec{q} \cdot \vec{v}$ be small compared to other terms in the denominator of (23.21), which means that

$$\frac{1}{\tau} - i\omega + i(\bar{n} + i\kappa) \frac{\omega v_F}{c} \approx \frac{1}{\tau} - i\omega \quad (23.23a)$$

The largest $\vec{q} \cdot \vec{v}$ can be is qv_F , where v_F is the largest velocity of an electron on the Fermi surface; the factor of $\partial f / \partial \mu$ restricts v to the Fermi surface. Use Eq. (23.22) for q .

$$\Rightarrow \frac{\bar{n}v_F}{c} \ll 1 \quad (23.23b)$$

and

$$\kappa\omega v_F \tau / c \ll 1 \quad \text{or equivalently} \quad l_T \ll \delta, \quad (23.23c)$$

where $l_T = v_F \tau$ is the electron mean free path, and

$$\delta \equiv \frac{c}{\kappa\omega} \quad (23.24)$$

is the *skin depth*, the characteristic distance that electromagnetic waves penetrate into a metal, according to Eq. (20.19). Condition (23.23b) is almost always satisfied, because Fermi velocities are two orders of magnitude smaller than the speed of light. Condition (23.23c) is satisfied by typical metals at room temperature, where mean free paths are on the order of 100 Å and where for frequencies up to the cutoff in (23.1) the skin depth is 1000 Å or greater, as will be shown below. It is violated by pure metals at temperatures on the order of 10 K, because the mean free path can rise as high as 10^{-2} cm.

Assuming that the conditions of (23.23) hold, σ does not depend upon $\vec{q}$. In an isotropic solid or a cubic crystal, the dielectric tensor can be treated as a scalar and equals

$$\epsilon = 1 - \frac{\omega_p^2}{\omega(\omega + i/\tau)} \quad (23.25)$$

with

$$\omega_p^2 = \frac{4\pi n e^2}{m_{\text{opt}}}, \quad (23.26)$$

the optical mass m_{opt} being defined by

$$\frac{1}{m_{\text{opt}}} = \frac{1}{m} \frac{\int [d\vec{k}] \frac{\partial f}{\partial \mu} m v_x^2}{\int [d\vec{k}] f_k} = \int \frac{d\Sigma}{12\pi^3 \hbar n} v. \quad (23.27)$$

Averages of v_x^2 equal averages of $v^2/3$. Denominator equals n .

The Drude results (20.31) and (20.32) have emerged unchanged from this analysis, except that the electron mass is replaced by the effective mass m_{opt} resulting from an average over the Fermi surface.

23.2.1 Anomalous Skin Effect

When the conditions of (23.23) do not hold, determining the dispersion relation for a metal rapidly becomes quite complicated. In the limit where the second of the conditions is reversed and the skin depth δ is much smaller than the mean free path, the complications are worth pursuing slightly further. The integral appearing in Eq. (23.21) changes from an average over the entire Fermi surface into an integral very sharply peaked on a line running around the Fermi surface. For this reason, the *anomalous skin effect* can be used to measure geometrical properties of the Fermi surface, and Pippard (1957) employed it to obtain the first experimental determination of the Fermi surface of copper.

Pippard's experiments were carried out at frequencies $\omega \approx 10^{11}$ Hz and at temperatures on the order of 10 K, where the relaxation time τ in copper rises from its room temperature value of 10^{-14} s to 10^{-10} s. The dielectric tensor changes very rapidly in this frequency regime, so one cannot simply speak of an index of refraction, but roughly speaking $\bar{n} \sim 10^4$, so that $q \sim \bar{n}\omega/c \sim 10^5 \text{ cm}^{-1}$. Because the Fermi velocity of copper is around $1.5 \cdot 10^8 \text{ cm/sec}$ it follows that $\omega \ll qv_F$, and ω can be neglected in the denominator of (23.21). The evaluation of Eq. (23.21) continues with the observation that because $\tau v_F q \gg 1$, the integral will be dominated by occasions where $\vec{q} \cdot \vec{v}_F = 0$ —that is, where the wave vector $\vec{q}$ is perpendicular to the direction of electron propagation $\vec{v}$, as shown in Figure 23.3. As radiation passes through the surface of the metal, it decays rapidly. Because the mean free path of electrons is much larger than the skin depth, electrons traveling parallel to the surface and perpendicular to $\vec{q}$ are excited into large amplitude oscillations.

To estimate the value of (23.20), suppose that radiation is arriving along $\vec{q} = q\hat{z}$, that it is polarized along x , and that the portion of the Fermi surface where $\vec{v} \cdot \vec{q} = 0$ can be approximated by two radii of curvature, $\mathcal{R}_\phi$ and $\mathcal{R}_\theta$, describing the curvature along the θ and ϕ directions depicted in Figure 23.3. Then

$$d\Sigma \approx \mathcal{R}_\phi \mathcal{R}_\theta d\theta d\phi, \quad v_x \approx v_F \cos \phi \quad (23.28)$$

and

$$\sigma_{xx} = e^2 \int \frac{\mathcal{R}_\phi \mathcal{R}_\theta d\theta d\phi}{4\pi^3 \hbar v_F} \frac{v_F^2 \cos^2 \phi}{1/\tau + iqv_F \theta} \quad (23.29)$$

Replacing $\vec{q} \cdot \vec{v}_F$ by $qv_F \theta$ is fine because the integrand drops off very rapidly away from $\theta = 0$.

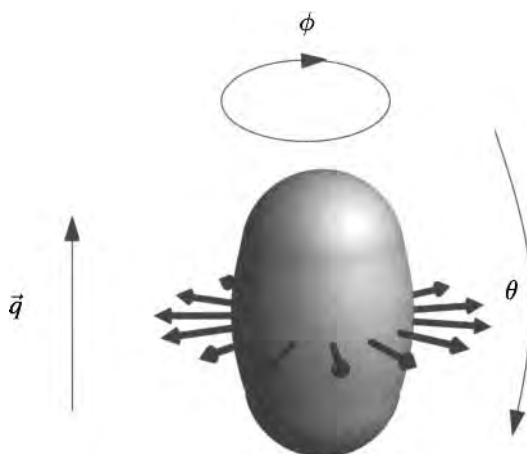


Figure 23.3. The electrons responsible for the anomalous skin effect have velocities perpendicular to the incoming radiation $\vec{q}$. Recall from Section 7.2.5, and Problem 1 in Chapter 7, that electron velocity vectors are normal to the Fermi surface.

$$= \frac{e^2}{4\pi\hbar q} \mathcal{R}_\phi \mathcal{R}_\theta. \quad (23.30)$$

Extending the limits of θ integration from $-\infty$ to ∞ is also fine because $qv_F\tau \gg 1$.

Without following the analysis further, it is clear that the relaxation time τ and Fermi velocity v_F have dropped out of the expression for the conductivity, which is governed by the curvature of the Fermi surface. Detailed calculation of absorption requires studying a highly dispersive wave as it enters the surface of the metal, and it will not be pursued further here. However, the geometrical information in Eq. (23.30) is enough to determine shapes of Fermi surfaces, and it was the first method used to obtain Figure 16.12.

23.3 Plasmons

The plasma frequency ω_p in Figure 23.2 where $\epsilon = 0$, $\bar{n} = \kappa$, and the metal passes from reflecting to transparent is worth examining further. Oscillations at frequency ω_p are only the longest wavelength limit of a whole family of oscillating modes. The general condition for a longitudinal propagating mode is $\epsilon(\vec{q}, \omega) = 0$, and the excitations obeying this condition above the plasma frequency are called *plasmons*. To calculate their properties one can turn to the dielectric function of the electron gas defined by Eqs. (20.84) and (20.77).

First look for a resonance at small $\vec{q}$. One can rewrite the susceptibility χ_c as

$$\chi_c = \frac{e^2}{\hbar v} \sum_{\vec{k}} f_{\vec{k}} \left[\frac{1}{\omega_{\vec{k}} - \omega_{\vec{q}+\vec{k}} - \omega} + \frac{1}{\omega_{\vec{k}} - \omega_{\vec{q}+\vec{k}} + \omega} \right] \quad (23.31)$$

$$= \frac{e^2}{\hbar v} \sum_{\vec{k}} \frac{2f_{\vec{k}}(\omega_{\vec{k}} - \omega_{\vec{k}+\vec{q}})}{(\omega_{\vec{k}} - \omega_{\vec{k}+\vec{q}})^2 - \omega^2} \quad (23.32)$$

$$= \frac{e^2}{\mathcal{V}} \sum_{\vec{k}} \frac{2f_{\vec{k}} \left[\frac{q^2}{2m} + \frac{\vec{q} \cdot \vec{k}}{m} \right]}{\omega^2 - \hbar^2 \left[\frac{\vec{q} \cdot \vec{k}}{m} + \frac{q^2}{2m} \right]^2}. \quad (23.33)$$

Because the goal is to find long-wavelength propagating modes in the vicinity of the plasma frequency ω_p , it is logical to treat $\vec{q}$ as small and expand (23.33) in powers of $\vec{q}$. Keeping the first two orders gives

$$\chi_c(\vec{q}, \omega) \approx \frac{e^2}{\mathcal{V}} \sum_{\vec{k}} \frac{f_{\vec{k}}}{\omega^2} \frac{q^2}{m} \left[1 + \frac{3(\vec{q} \cdot \vec{k})^2 \hbar^2}{m^2 \omega^2} \right]. \quad (23.34)$$

Carrying out an angular average, the factor of 3 disappears, leaving

$$\chi_c(\vec{q}, \omega) = \frac{e^2}{\mathcal{V}} \sum_{\vec{k}} \frac{f_{\vec{k}}}{\omega^2} \frac{q^2}{m} \left[1 + \frac{(qk)^2 \hbar^2}{m^2 \omega^2} \right]. \quad (23.35)$$

The sum over $\vec{k}$ includes a sum over all angles of $\vec{k}$, so the result cannot change if terms are replaced by their angular average.

Recall that

$$N = \sum_{\vec{k}\sigma} f_{\vec{k}} = \mathcal{V} \int \frac{dk}{4\pi^3} 4\pi k^2 f_{\vec{k}}. \quad (23.36)$$

So one must have that

$$\sum_{\vec{k}\sigma} f_{\vec{k}} k^2 = \frac{3}{5} N k_F^2. \quad (23.37)$$

Using Eqs. (23.35) and (23.37) in Eq. (20.84) gives

$$\epsilon(\vec{q}, \omega) = 1 - \frac{4\pi n e^2}{m\omega^2} \left[1 + \frac{3}{5} \frac{\hbar^2 k_F^2 q^2}{m^2 \omega^2} \right]. \quad (23.38)$$

So longitudinal modes propagate when $\epsilon = 0$, which means

$$1 = \frac{\omega_p^2}{\omega^2} \left[1 + \frac{3}{5} \frac{\hbar^2 k_F^2 q^2}{m^2 \omega^2} \right] \quad (23.39)$$

$$\Rightarrow \omega^2 = \omega_p^2 + \frac{6}{5} \frac{\mathcal{E}_F q^2}{m}. \quad (23.40)$$

Equation (23.40) is the dispersion relation for plasmons.

23.3.1 Experimental Observation of Plasmons

Plasmons are fairly long-lived until they achieve a wave vector at which it is possible for a plasmon to transfer some of its energy to a single moving electron. At this point, the plasma waves decay rapidly, in a phenomenon known as *Landau damping*. The condition for Landau damping may be obtained by examining when it becomes possible for the denominator in Eq. (23.33) to vanish with ω and q related by Eq. (23.40), and $\vec{k}$ within the Fermi surface.

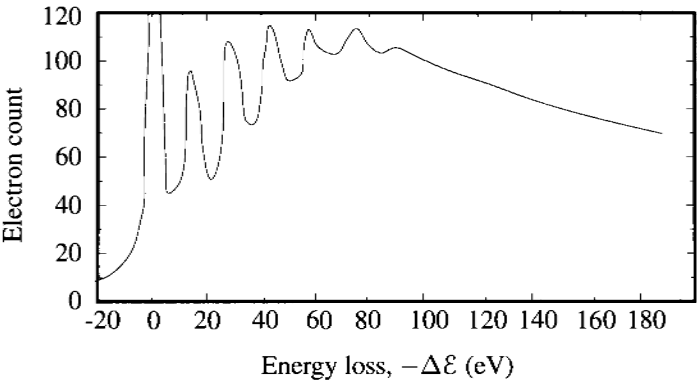


Figure 23.4. Number of scattered electrons as a function of energy loss. The peaks correspond to creation of zero, one, two, or more plasmons. [Source: Lang (1948), p. 241.]

The classic experiments measuring such plasma oscillations in metals were performed simultaneously by Ruthemann (1948) and Lang (1948). A monoenergetic electron beam with an energy in the kilovolt range was passed through a 500-Å aluminum film. In aluminum $\hbar\omega_p$ is 15.8 eV, and collision events involving the creation of one or a few plasmons appear as energy loss peaks in multiples of this quantum after the electron beam exits the film. A sample of such data appears in Figure 23.4.

Just as in the cases of neutron scattering for the measurement of phonon dispersion relations, or photoemission for the measurement of electronic band structure, the dispersion of plasmons can be measured through an inelastic scattering experi-

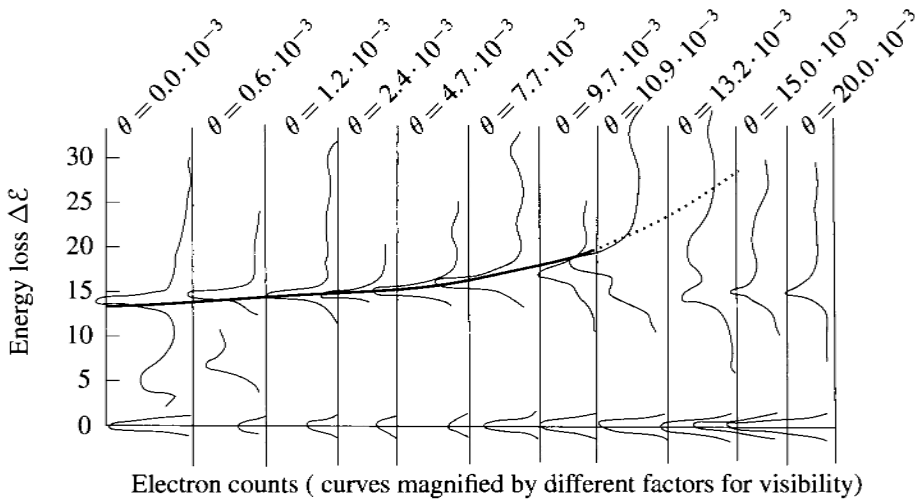


Figure 23.5. Energy loss versus number of scattered electrons for numerous scattering angles θ . Only a single plasmon peak is visible, and it disappears when the conditions for Landau damping are met. The line through the peaks shows the plasmon dispersion relation $\hbar\omega(\theta)$. [Source: Kunz (1962) p. 59.]

Table 23.1. Comparison of measured and predicted plasmon dispersion relations

Element	Be	Al	Mg	Sb	Na
α_{pl} [from Eq. (23.43)]	0.47	0.44	0.39	0.44	0.32
α_{pl} (experiment)	0.42	0.35	0.39	0.37	0.29

Source: Platzman and Wolff (1973) p. 71.

ment. Plasmons are excited by electrons injected into samples. Inelastic scattering with electrons is called *electron energy loss spectroscopy* (EELS), and is reviewed by Schnatterly (1979). Equations (13.93) can be used to analyze data of the type shown in Figure 23.5.

In particular, suppose that an electron enters a sample, losing energy $\Delta\mathcal{E}$, and changing direction from $\vec{k}$ to $\vec{k}'$, differing by a small angle θ . Then conservation of energy and momentum require that

$$\hbar\omega(\vec{k} - \vec{k}') = \Delta\mathcal{E} \quad (23.41)$$

$$\Rightarrow \hbar\omega(2k \sin \theta/2) \approx \hbar\omega_p + \alpha_{\text{pl}} \frac{\hbar^2 k^2}{m} \theta^2 \quad (23.42)$$

Use Eq. (23.40), the fact that θ is small and Eq. (3.8); the change in electron angle θ is twice the Bragg angle of Eq. (3.8).

where

$$\alpha_{\text{pl}} = \frac{3}{5} \frac{\hbar^2 k_F^2}{2m\hbar\omega_p}. \quad (23.43)$$

The results are satisfactory, as shown in Table 23.1.

23.4 Interband Transitions

As the frequency of incoming photons increases toward 1 eV, so does the probability that light will induce transitions where electrons hop between bands. Actually, there is no reason in a metal why the such transitions cannot occur for arbitrarily small photon energies, but the minimum jump tends to be on the order of 1 eV. The dominant optical transition in such a case is described by the same equation, Eq. (21.14), that was employed to discuss direct transitions in semiconductors.

Example: Sodium. Interband transitions in the alkali metals are particularly simple, because the conduction band electrons behave very nearly like free non-interacting electrons, and have energies well above any of the core electrons. The Fermi surface is nearly spherical, and deviations from the completely free electron model can be attributed to a weak pseudopotential. These claims result from band structure calculations, and are also in excellent agreement with the optical measurements to be discussed below, as well as with de Haas–van Alphen measurements of the type discussed in Section 16.5.2.

All of the alkali metals adopt the bcc crystal structure in their ground state. The relation of the Fermi surface to the edges of the Brillouin zone is given approximately by the upper left image in Figure 8.7. The closest approach to the zone edge

is at point N , labeled in Figure 7.9, which is at distance $2\pi/\sqrt{2}a = 4.44/a$ from the origin, while the Fermi surface is at distance $[6\pi^2]^{1/3}/a = 3.89/a$. Figure 23.6 shows the computed band structure of sodium along the line $\Gamma - N$. At the location of the Fermi surface, the energy bands are extremely close to the free electron values, and therefore the wave functions are accurately given by Eq. (8.14).

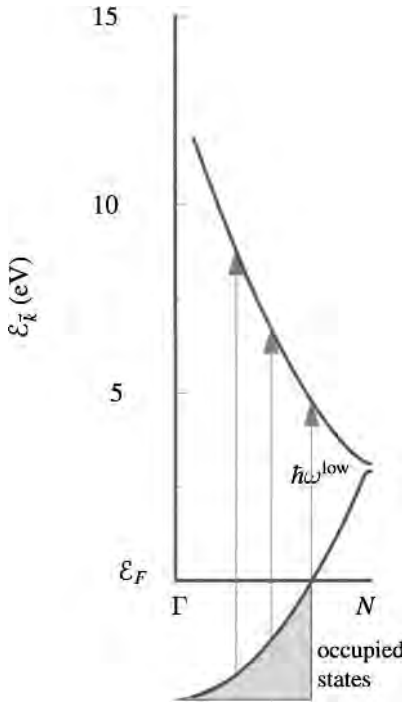


Figure 23.6. The sodium electron bands are nearly indistinguishable from free electron bands except near zone edges, and the Fermi level lies sufficient distance away that the nearly free electron approximation is excellent. The smallest interband transition, $\hbar\omega^{\text{low}}$, takes an electron from the Fermi surface up to the next band.

In order to evaluate Eq. (21.16), one needs to find the matrix elements

$$\langle n_1 \vec{k} | \hat{P}_\alpha | n_2 \vec{k} \rangle. \quad (23.44)$$

There is only one occupied band n_1 , but there are 12 reciprocal lattice vectors equivalent to (110) providing upper bands n_2 that must be considered. Their contributions are simply additive, so it is enough to deal with one at a time. According to (8.15), which dealt with two bands at a time in the extended zone representation, the wave functions of electrons in the lower and upper bands are

$$\psi_{\vec{k}}^{\text{low}}(\vec{r}) \approx \frac{1}{\sqrt{V}} \left[e^{i\vec{k} \cdot \vec{r}} + \frac{e^{i(\vec{k}-\vec{K}) \cdot \vec{r}} U_{-\vec{K}}}{\mathcal{E}_{\vec{k}}^0 - \mathcal{E}_{\vec{k}-\vec{K}}^0} \right] \quad (23.45)$$

This is the wave function corresponding to the lower branch in Figure 23.6. In employing Eq. (8.15), $\vec{K}$ has been replaced by $-\vec{K}$.

$$\psi_{\vec{k}}^{\text{high}}(\vec{r}) \approx \frac{1}{\sqrt{V}} \left[e^{i(\vec{k}-\vec{K}) \cdot \vec{r}} + \frac{e^{i\vec{k} \cdot \vec{r}} U_{\vec{K}}}{\mathcal{E}_{\vec{k}-\vec{K}}^0 - \mathcal{E}_{\vec{k}}^0} \right]. \quad (23.46)$$

This is the wave function corresponding to the upper branch in Figure 23.6.

Most details are left to Problem 2. In brief, (20.70) becomes

$$\text{Re}[\sigma_{\alpha\beta}](\omega) = \frac{\pi}{\omega} \frac{e^2 \hbar^2}{m^2} \frac{1}{\mathcal{V}} \sum_{\vec{k}, \vec{K} \in \langle 110 \rangle} f_{\vec{k}} \frac{|U_{\vec{K}}|^2 K_{\alpha} K_{\beta}}{(\mathcal{E}_{\vec{k}-\vec{K}}^0 - \mathcal{E}_{\vec{k}}^0)^2} \delta(\mathcal{E}_{\vec{k}-\vec{K}}^0 - \mathcal{E}_{\vec{k}}^0 - \hbar\omega) \quad (23.47)$$

$\vec{k}$ is summed over the Brillouin zone, and $\vec{K}$ ranges over the vectors equivalent to $\langle 110 \rangle$ under cubic symmetry operations.

$$\Rightarrow \sigma(\omega) = \frac{4e^2 \pi}{m^2 \omega^3} K^2 |U_{\vec{K}}|^2 D_j(\hbar\omega), \quad (23.48)$$

where the joint density of states D_j defined in Eq. (21.17) takes the form

$$D_j(\hbar\omega) = \frac{1}{\mathcal{V}} \sum_{\vec{k}, \vec{K} \in \langle 110 \rangle} f_{\vec{k}} \delta(\mathcal{E}_{\vec{k}-\vec{K}}^0 - \mathcal{E}_{\vec{k}}^0 - \hbar\omega) \quad (23.49)$$

$$= \frac{m^3}{4\pi^2 \hbar^4 K^3} (\omega^{\text{high}} - \omega)(\omega - \omega^{\text{low}}) \quad \begin{array}{l} \sigma \text{ vanishes for } \omega > \omega^{\text{high}} \text{ and} \\ \omega < \omega^{\text{low}}. \end{array} \quad (23.50)$$

with

$$\omega^{\text{high}} = \frac{\hbar^2 K(K + 2k_F)}{2\hbar m} \quad \omega^{\text{low}} = \frac{\hbar^2 K(K - 2k_F)}{2\hbar m}. \quad (23.51)$$

Figure 23.7 shows experimental measurements of absorption in the alkali metals. The onset of absorption is in accord with the theoretical results, and it can be used to estimate the size of the pseudopotential $U_{\vec{K}}$, as shown in Problem 2.

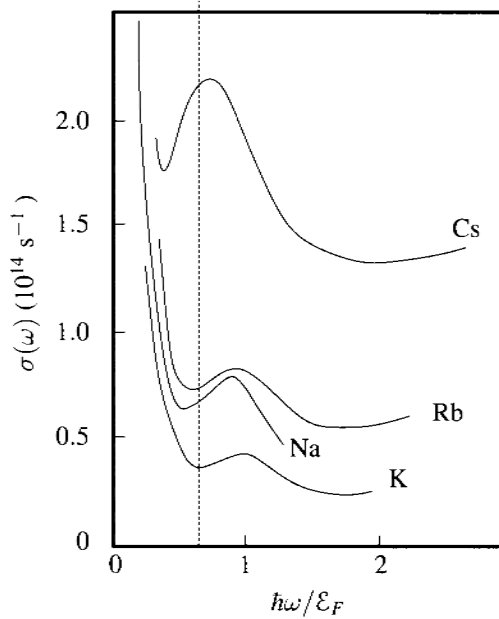


Figure 23.7. Optical absorption of alkali metals, showing quadratic peak due to onset of interband absorption. The minimum in absorption at ω^{high} lies at higher frequencies than those shown and it would likely be masked by transitions between additional bands. [Source: Smith (1970), p. 3664.]

Noble Metals. The noble metals differ from the alkalis because the filled d band lies comparatively close to the Fermi surface, as shown in Figure 10.7. This fact has two consequences. First, it means that in the frequency range where the dielectric constant is dominated by the s electrons, ϵ^∞ is quite large, on the order of 7, because the d electrons provide a highly polarizable environment, roughly as depicted in Figure 20.4. Second, the first interband transitions are not between electrons at the Fermi surface and unoccupied bands above. Instead, the lowest-energy jumps are between the d bands and the Fermi surface, as predicted by Figure 10.7(A). Experimental measurements of absorption in the three noble metals appear in Figure 23.8, showing that absorption begins at the expected frequencies.

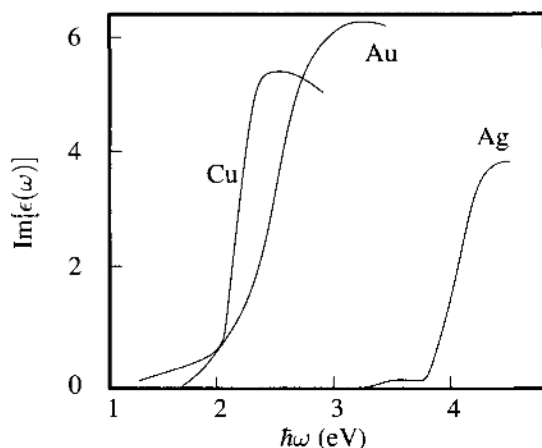


Figure 23.8. Imaginary part of the dielectric constant measured for copper, silver and gold. The absorption edge occurs close to the energy of around 2 eV predicted for copper in Figure 10.7(A), although it is not completely sharp. [Data of Thèye (1968), published by Abelès (1972) p. 138.]

23.5 Brillouin and Raman Scattering

Electromagnetic waves impinging upon samples have so far been presumed to pass through with their frequency unchanged. Brillouin (1922) proposed that photons entering a sample with energy $\hbar\omega_0$ could leave with energy $\hbar(\omega_0 - \omega_1)$ if an excitation of energy $\hbar\omega_1$ were created in the process. Raman (1928) was the first to measure this phenomenon directly. The excitations responsible for changing the frequency of light can be phonons, magnons, or any of the more elaborate quasi-particles such as excitons or polaritons.

Just as for inelastic neutron scattering, two conservation laws control the inelastic scattering of light, and photons can either lose or gain energy depending upon whether an excitation is created or destroyed. Conservation of energy obviously requires that the energy change of the excitation match the energy change of the light. In the case where an excitation is created, conservation of crystal momentum requires that

$$\vec{k}_f = \vec{k}_0 - \vec{k}_1 \quad \begin{array}{l} \vec{k}_f \text{ is the final wave vector of the light, } \vec{k}_0 \text{ is the} \\ \text{initial wave vector, and } \vec{k}_1 \text{ is the wave vector of} \\ \text{the excitation.} \end{array} \quad (23.52)$$

In a medium with index of refraction $\bar{n}$, light obeys the dispersion relation $\omega = ck/\bar{n}$, so the two conservation laws can be combined to read

$$\frac{c}{\bar{n}}(k_f - k_0) = \mp \omega_1 \left(\mp (\vec{k}_f - \vec{k}_0) \right). \quad (23.53)$$

The minus sign creates an excitation, while the positive sign destroys one. $\omega_1(\vec{k})$ is the dispersion relation of the excitation. Compare with Eq. (13.93).

Light whose frequency is reduced is *Stokes scattered* light, while light whose frequency increases is *anti-Stokes scattered* light, and the frequency change is called the *Raman shift*.

Brillouin's theory concerned the scattering of light from sound waves, so when inelastic light scattering creates and destroys acoustic phonons it is known as *Brillouin scattering*, while Raman's experiments involved optical phonons, and inelastic scattering from these excitations is called *Raman scattering*. Because of the high photon fluxes available from synchrotrons, X-rays can be employed as well as visible light, leading to the technique of *inelastic X-ray scattering*.

Referring to Figure 20.1, the wave number of visible photons is several orders of magnitude smaller than the wave number of phonons near the zone edge. For this reason, Raman and Brillouin scattering are only able to look at phonons near the zone center. Inelastic X-ray scattering, however, is able to explore phonon dispersion relations through the entire Brillouin zone.

23.5.1 Brillouin Scattering

In the case of acoustic phonons,

$$\omega_1 = c_p k. \quad \begin{array}{l} c_p \text{ is the sound speed of a transverse or longitudinal phonon.} \end{array} \quad (23.54)$$

Adopting the case of Stokes scattering—the minus sign in Eq. (23.53)—and taking θ to be the angle between incoming and outgoing light, one has

$$(k_f - k_0) = -\frac{\bar{n}c_p}{c} \sqrt{k_f^2 + k_0^2 - 2k_f k_0 \cos \theta} \quad (23.55)$$

$$\Rightarrow k_0 - k_f \approx k_0 \frac{2\bar{n}c_p}{c} \sqrt{\frac{1 - \cos \theta}{2}}. \quad \begin{array}{l} \text{Because } c_p/c \ll 1, k_f \text{ and } k_0 \text{ can be set equal to leading approximation on the right-hand side.} \end{array} \quad (23.56)$$

$$\Rightarrow \omega_0 - \omega_f = \frac{2\bar{n}\omega_0 c_p}{c} \sin \theta/2 \quad (23.57)$$

The data shown in Figure 23.9 show Brillouin scattering from longitudinal and transverse phonons in germanium. The phonon frequencies were already known from acoustic measurements, so the data were used to extract the complex dielectric constant $\bar{n}$ of germanium, with the width of the peaks providing a measure of the extinction coefficient κ .

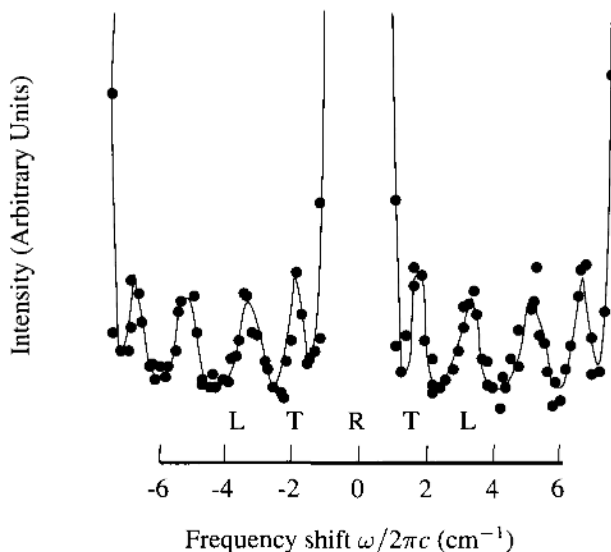


Figure 23.9. Brillouin scattering from the (111) surface of germanium around a wavelength of 6328 Å. The transverse (T) and longitudinal (L) frequencies, as well as multiples of the longitudinal and transverse phonon frequencies due to multiple phonon absorption or emission are visible around the central Rayleigh (R) peak. [Source: Sandercock (1972), p. 239.]

23.5.2 Raman Scattering

In scattering off optical phonons, the frequency ω_1 is nearly constant near the zone center; Raman scattering is frequently used to determine this constant. A more elaborate use of the technique is the measurement of the polariton dispersion relation shown in Figure 23.10. Because large variations in this dispersion relation occur for very small wave numbers, Raman scattering has an advantage over neutron scattering in this situation.

23.5.3 Inelastic X-Ray Scattering

The feasibility of using inelastic X-ray scattering to measure phonon spectra was demonstrated by Dorner et al. (1987), and their measurements of the phonon dispersion relation of beryllium appear in Figure 23.11.

23.6 Photoemission

23.6.1 Measurement of Work Functions

As photon energies rise toward 4 eV, a new sort of transition becomes possible, one in which a photon not only excites an electron to a higher energy band, but ejects it from the sample altogether. The physics governing the ejection of electrons by photons is essentially the same that governs ejection by elevated temperature, as in the Richardson–Dushman equation, Eq. (19.8). Because a photon of frequency ω

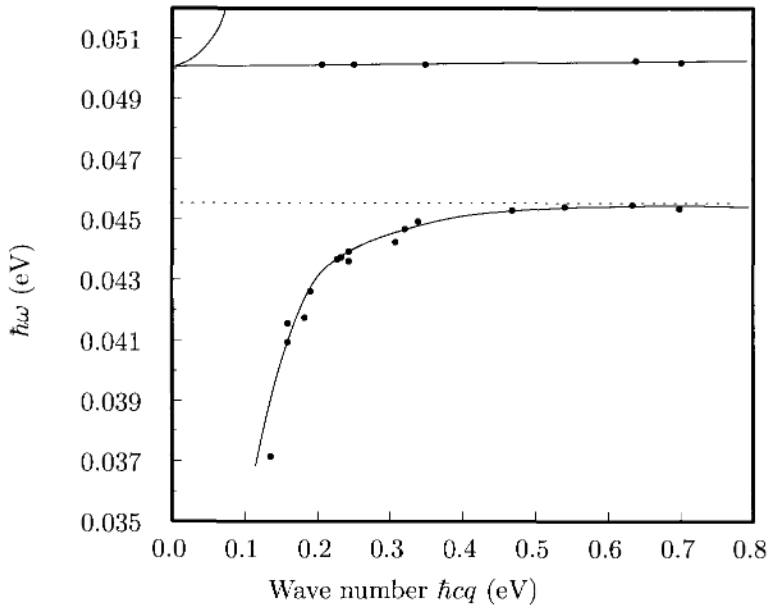


Figure 23.10. Dispersion relation of polaritons in GaP measured with Raman scattering. The data are taken in (100) and (111) directions, and also in polycrystals. To scan through the dispersion relation, one frequency of incoming laser light was used, but outgoing light intensity was measured as a function of energy for a range of angles θ . Compare with Figure 22.5, where a similar dispersion relation was obtained indirectly from absorption measurements. [Source: Henry and Hopfield (1965), p. 965.]

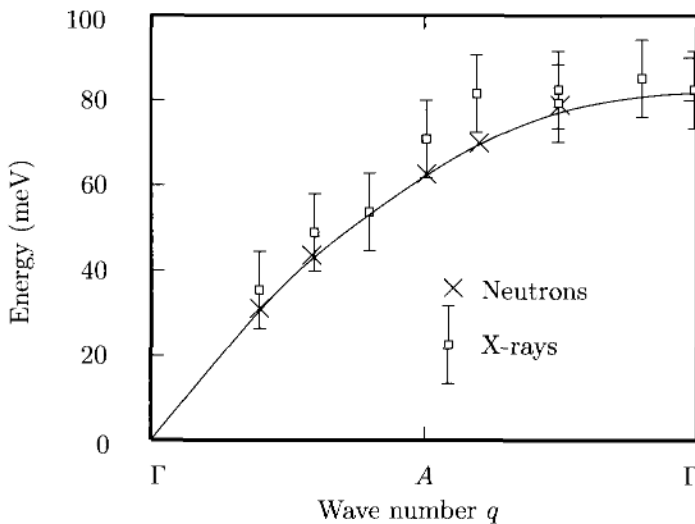


Figure 23.11. Dispersion relation of longitudinal phonons in beryllium along the c axis, comparing inelastic X-ray scattering with neutron scattering. Prior to this experiment, such measurements with X-rays had been thought impractical; the situation changed because of the large photon flux possible at a synchrotron. [Source: Dorner et al. (1987), p. 182.]

can supply energy $\hbar\omega$ to an electron, the effect of light upon a sample is to reduce the work function to $\phi - \hbar\omega$, and when $\hbar\omega > \phi$, electrons are free to flow out, in proportion to the incoming photon density. A plot of photoelectron current versus photon energy appears in Figure 23.12, and a table of work functions measured in this way appears in Table 23.2.

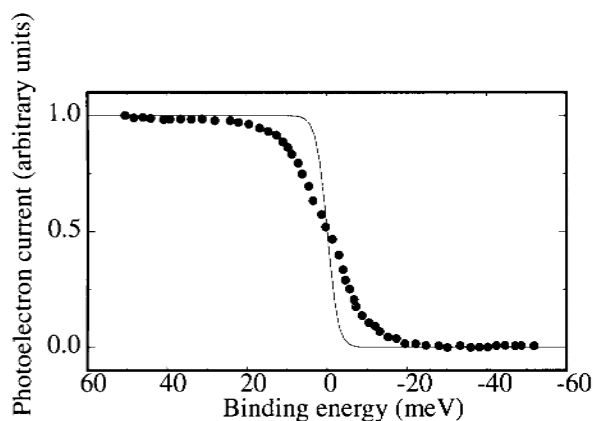


Figure 23.12. Electron current versus photon energy relative to energy of work function of silver at 15 K. The measurement is sensitive enough that the profile of the Fermi function is visible. The data are somewhat broader than the appropriate Fermi function; the deviation is a measure of divergences in the experimental system from the one-electron picture. [Source: Patthey et al. (1990), p. 8872.]

Table 23.2. Work functions of selected metals and compounds

Compound	Surface	ϕ (eV)	Compound	Surface	ϕ (eV)
Ag	(100)	4.64	Na	(110)	2.9
	(110)	4.52	Nb	(100)	4.02
	(111)	4.74		(110)	4.87
Al	(100)	4.20		(111)	4.36
	(110)	4.06	Ni	(100)	5.22
	(111)	4.26		(110)	5.04
Au	(100)	5.47		(111)	5.35
	(110)	5.37	Pt	(100)	5.84
	(111)	5.31		(111) 2×1	4.85
Be	(0001)	5.1		(111) 7×7	4.50
Cu	(100)	5.10	W	(100) 2×1	4.87
	(110)	4.48		(100)	4.63
	(111)	4.94		(110)	5.25
Fe	(100)	4.67		(111)	4.47
Ge	(111) 2×1	4.68	SiC	(0001)	4.6
	(111) 2×8	4.53	AlN	(100)	5.35
K	(110)	2.39	GaAs	(110)	5.56
Mg	(100)	3.71	GaSb	(110)	4.91
Mo	(100)	4.53	InP	(110)	5.85
	(110)	4.95			
	(111)	4.55			

Values are obtained mainly by photoemission, but also from thermal emission using Eq. (19.9), and by measuring currents emitted under strong electric fields. Source: Landolt and Börnstein (New Series) vol. 17 and Hölzl and Schulte (1979), p. 86.

23.6.2 Angle-Resolved Photoemission

Because of the close association of photoemission experiments with the work function and because the work function is connected with transitions between solid and vacuum, for a long time it was thought that photoemission could measure only properties of solid surfaces. It can be used very effectively for that purpose, but has even more important application in measuring bulk energy bands. *Angle-resolved photoemission spectroscopy* (ARPES) has become the most important experimental tool for probing the electronic properties of solids, and makes it possible to map out energy bands directly.

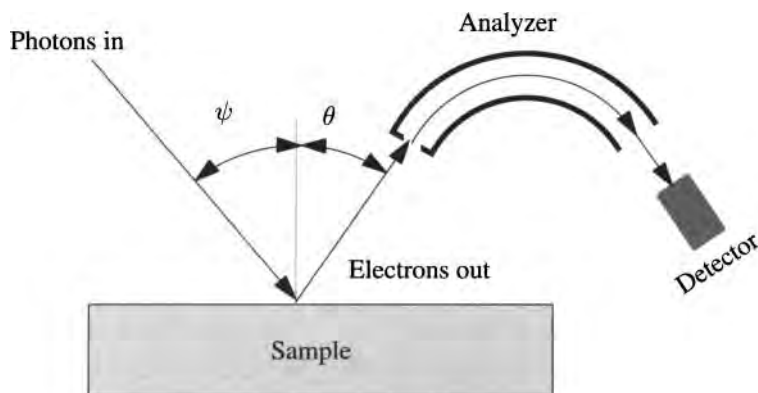


Figure 23.13. Schematic view of angle-resolved photoemission experiment. Photons are incident upon a surface at angle ψ , and electrons are collected at angle θ . Passing the electrons through a curved chamber with an electric field that sends the electrons in a curved path, only those in a narrow range of kinetic energies survive to impact upon the detector. Varying the electric field allows a measurement of photoelectron current as a function of kinetic energy.

A schematic view of the experiment is presented in Figure 23.13. Photons are directed toward the surface of a sample at an angle ψ , electrons emitted along angle θ are collected, and the amplitude of the photocurrent is measured as a function of the electrons' kinetic energy. There is a wide range of variables to vary; the angles of incoming photons and outgoing electrons, the energies of the incoming photons and outgoing electrons, and the Miller index of the crystal surface. A single experiment will rarely vary all of these, and a characteristic set of data appears in Figure 23.14, where both electrons and photons travel normal to the crystal, and photocurrent is measured versus final electron energy for a range of incident photon energies.

To understand the significance of this technique, it is useful to make a comparison with two other types of experiment, optical absorption, and neutron scattering. According to Eq. (20.70), optical absorption is produced by sums over electron transitions between a ground state and all final states compatible with energy conservation. Angle-resolved photoemission breaks the sum into constitutive components by finding the transition rate to states where an outgoing electron has known

energy and momentum. Ideally one would like to think of the experiment as being entirely analogous to neutron scattering, with the photon taking the place of the incoming neutron, while the outgoing electron takes the place of the outgoing neutron. Just as neutron scattering contained enough information to obtain phonon dispersion relations, angle-resolved photoemission should contain enough information to obtain electron band dispersion relations. In the spirit of Section 13.4.1, one can write down the consequences of energy and momentum conservation. Energy conservation demands

$$\phi + \mathcal{E}_{\text{kin}} - (-\mathcal{E}_B) = \hbar\omega, \quad (23.58)$$

where $\mathcal{E}_B$, the *binding energy* is the negative of the electron's energy before impact by the photon, and $\phi + \mathcal{E}_{\text{kin}}$, work function plus kinetic energy, is its energy after being ejected. Because photon momentum is negligible, crystal momentum conservation would demand that the final wave vector $\vec{k}_f$ of the ejected electron be equal to its initial wave vector up to the inevitable addition or subtraction of a reciprocal lattice vector so as to lie in the first Brillouin zone. So Eq. (23.58) would become

$$\mathcal{E}_B(\vec{k}_{\text{final}}) = \hbar\omega - \phi - \mathcal{E}_{\text{kin}}, \quad \text{Map } \vec{k}_f \text{ back into the first Brillouin zone if necessary.} \quad (23.59)$$

and energy levels $\mathcal{E}_B(\vec{k}_{\text{final}})$ could be mapped out as a function of $\vec{k}_{\text{final}}$. In fact this is almost exactly how the experiments are interpreted, but there are considerable grounds for worry. The old view that photoemission should be tremendously sensitive to the surface was not all wrong, and the difficulty manifests itself in the fact that momentum of the electron in the direction perpendicular to the sample surface has no reason to be conserved. Right at the onset of photoemission this claim is very easy to understand. A photon travels into the sample and transfers energy to an electron that most likely already has some momentum $\hbar k_{\perp}$ in the direction of the surface. On its way out of the sample the electron decelerates, and finally emerges with almost no momentum at all, so $\hbar k_{\perp}$ is certainly not conserved. Momentum parallel to the sample surface is conserved because the symmetry that produces the conservation law still holds; the system remains invariant when translated through a lattice vector parallel to the surface. It clearly does not remain invariant when displaced perpendicular to the surface, so the conservation law in this direction is lost.

In principle the missing information about the original momentum of the electron can be recovered by conducting experiments with different crystal surfaces, and then combining the information from the multiple experiments. While such experiments have been done, there is an easier solution in practice. A classical particle passing rapidly through a short-ranged force loses very little momentum. To be more precise, a particle traveling along $\hat{x}$ through a potential that varies by amount ΔU in a short distance experiences a change in momentum

$$\Delta p = \int -\frac{\partial U}{\partial x} dt \approx -\frac{\Delta U}{v}, \quad \text{Assuming that the velocity } v \text{ can be taken nearly constant.} \quad (23.60)$$

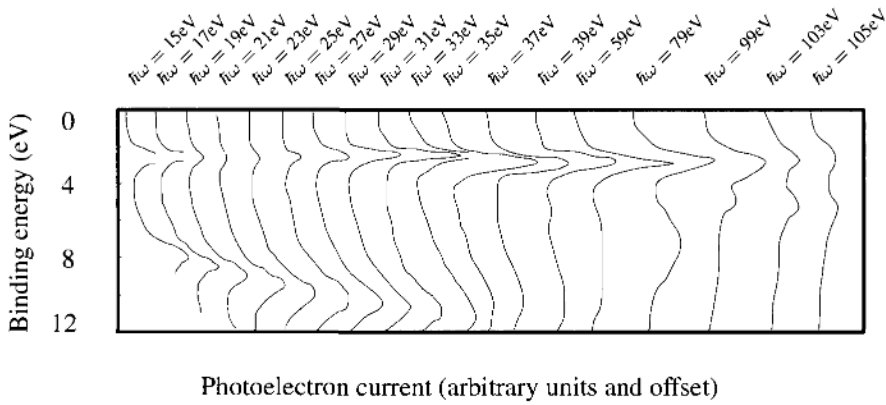


Figure 23.14. Characteristic raw data from a photoemission experiment, for photon injection and electron emission in beryllium along [0001]. The curves show photoelectron current as a function of energy for a large number of photon energies $\hbar\omega$. The curves are offset from one another horizontally so as all to be visible. The upper peak, with a binding energy of around 2.5 eV is independent of incident photon energy and is therefore likely to be a surface state. The lower peak is dispersive, and likely to correspond to a bulk band. [Source: Jensen et al. (1984), p. 5502.]

where v is its velocity, so the faster the particle moves the less momentum it loses. For this reason, angle-resolved photoemission spectroscopy is carried out with photons in the range of 10–1000 eV (*ultraviolet photoemission spectroscopy*, UPS) or above 1000 eV (*X-ray photoelectron spectroscopy*, XPS). A bit of uncertainty about $k_{\perp}$ remains, but can be resolved by identifying critical points where $\partial\mathcal{E}_B/\partial k$ vanishes and changes sign with zone edges.

Example: Beryllium. Radiation at the necessary frequencies and intensities became available with the advent of the synchrotron. Figure 23.14 illustrates the type of data that photoemission provides. Photons impact a beryllium surface along [0001], and electrons are detected returning in the same direction and sorted as a function of their kinetic energy. There are two separate bands visible in the figure. The upper one is nondispersive; the binding energy is independent of the incoming photon energy. Frequently, such nondispersive bands are surface states. The initial momentum $\hbar k_{\perp}$ perpendicular to the surface is guaranteed to be zero, and the momenta $\vec{k}_{\parallel}$ parallel to the surface can be measured without uncertainty (they too are zero in the present case). So the upper band is simply reporting the energy of a surface state at $\vec{k} = 0$. The lower band is dispersive, and therefore it results from a bulk electron energy band. To a first approximation, one can assume that the final momentum of the electron equals $\hbar k_{\perp}$, but in addition there are clues within the data that help improve upon this approximation. Notice that the lower band has maxima and minima. They should correspond to cases where $k_{\perp}$ reaches points of symmetry in the Brillouin zone. Referring to Figure 7.10, the minimum at $\hbar\omega = 35$ eV is point A, while the maximum at around 103 eV is probably Γ . In this fashion the data provide a direct measurement of $\mathcal{E}_{n\vec{k}}$ as a function of $\vec{k}$.

Semiconductors. Some of the most careful studies of this type have been carried out to investigate the band structures of silicon, germanium, and gallium arsenide. Figures 23.15 and 23.16 show theoretically computed band structures compared with photoemission data. In addition to mapping out bands below the Fermi surface, *inverse photoemission* can be used to map out bands above the Fermi surface. Inverse photoemission consists in passing electrons of known energy into a sample, and measuring the ejected photons. The electrons must go into unoccupied states, which is why the information provided by inverse photoemission is complementary to the information given by photoemission.

The theoretical band structures are best viewed as collaborations between theory and experiment. In order to ensure that band gaps agree with experiment, innovative terms carrying information about electron–electron interaction are added to the calculations. The band structures assembled in Section 10.4 already represent complicated calculations, but the one in Figures 23.16 and 23.15 are even more difficult to reproduce. The validation of the one-electron picture of electronic structure is nevertheless remarkable. The one-electron approximation has real experimental meaning, and electronic excitations of the solid behave as theory says they should.

There are many interesting complications associated with the interpretation of photoemission spectra. The intensity peaks always have a width of several electron volts, a width that is considerably greater than experimental resolution. The width is a result of *many-body effects*—that is, the scattering of electrons off of

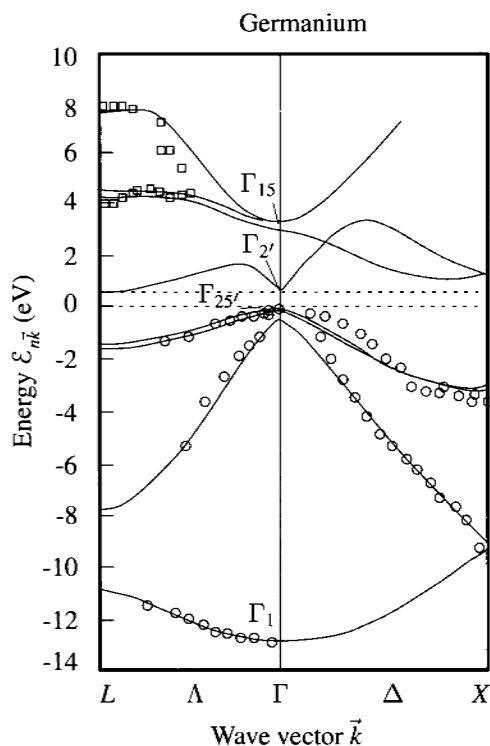


Figure 23.15. Theoretical energy bands of germanium compared with direct and inverse photoemission data. The theoretical calculations are due to Louie (1992), p. 79; they consist of density functional calculations supplemented with information from the theory of electron interaction. The photoemission experiments are due to Wachs et al. (1985) and Straub et al. (1986).

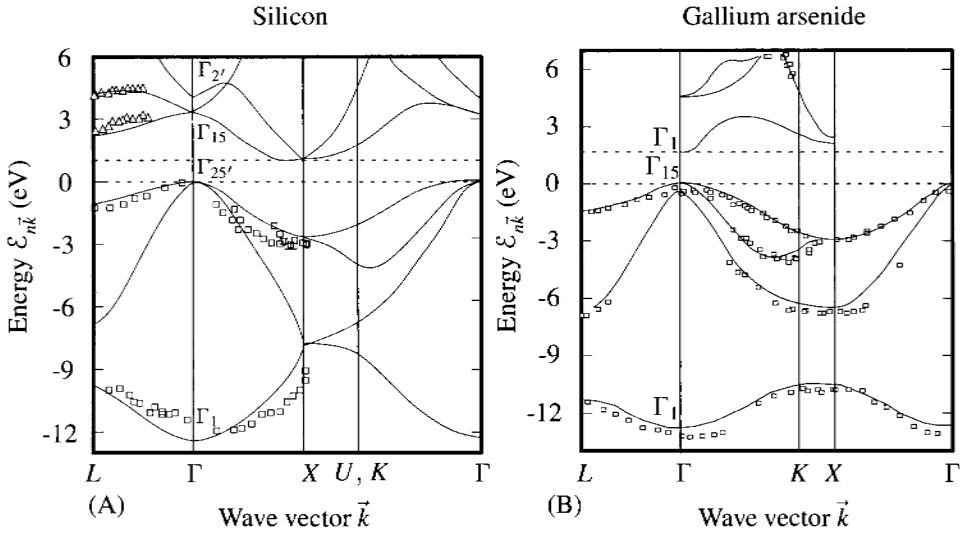


Figure 23.16. (A) Theoretical energy bands of silicon calculated with semiempirical pseudopotentials by Chelikowsky and Cohen (1976) compared with photoemission studies of Straub et al. (1986) and Rich et al. (1989). Notation for the symmetries of wave functions along Γ given in Table 7.4. (B) Theoretical energy bands of gallium arsenide calculated with semiempirical pseudopotentials by Pandey and Phillips (1974) compared with photoemission studies of Chiang et al. (1980) and Williams et al. (1986).

one another and the resulting decay of the single-particle states. There is much formalism to deal with this problem, but it is not easy to get practical information out of it. Matters are further complicated by the fact that interactions between surface and bulk states can move peaks around in a manner that falls outside the simple description used here.

23.6.3 Core-Level Photoemission and Charge-Transfer Insulators

Core-level photoemission is a curiously indirect technique. The incoming photons are much more energetic than in the photoemission experiments described so far, with typical photon energies on the order of 1000 eV, which places them in the X-ray range and giving the method also the name *X-ray photoelectron spectroscopy*. The requirement of a powerful coherent source of X-rays means that experiments of this type did not become possible before the availability of synchrotron radiation (Section 3.4.2.) The photons enter the atom, bypass the valence electrons, and knock out a core electron whose binding energy is approximately equal to that of the incoming photon. Nevertheless, the experiment is not conducted in the expectation of learning anything about the core electrons. Valence electrons still provide the object of study. Knocking out a core electron has the effect of suddenly adding a positive charge e to the core of an atom, as if an extra proton had somehow been injected into the nucleus, and the interest of the experiment lies in discovering how the valence electrons nearby will respond. The spatially localized nature of the per-

turbation naturally leads to information about the solid with a localized character. The type of information that can be extracted and the way it is interpreted is best illustrated with a particular example.

Transition Metal Oxides. Even during the first days that the single-electron band theory of solids was being constructed, it was realized that there existed solids that violated its predictions in a qualitative way. The prototypical example is NiO, as discussed by Mott (1949). The transition metal oxides, from VO through ZnO, have many features in common. Most of them adopt the NaCl structure (Figure 2.7). Density functional theory predicts them to be metals. In fact, they are all insulators, with an optical band gap of 1–4 eV, and all antiferromagnetic (Figures 3.14 and 24.6). The difficulty presented by these compounds is particularly evident for VO, MnO, and CoO, which have an odd number of electrons in each unit cell. The valence bands are not filled, so one can state with certainty in the single-electron approximation that they must be metals. Optical absorption of CoO appears in Figure 23.17, which clearly shows an optical gap of 4 eV. In the case of TiO, CrO, Fe, and NiO, the number of electrons per unit cell is even, so these compounds could in principle be insulators. However, the valence bands are built from the $3d4s$ shells, which, after lending two electrons to form O^{2-} , are incomplete. Detailed calculations almost always find these compounds to be metals. Density functional calculations for NiO do find it to be insulating once antiferromagnetic ground states are explored, but the calculated gap is much smaller than the one measured experimentally.

Among these transition metal compounds, CuO is both the most complicated and the most interesting. It stands alone in adopting a complicated crystal structure, which is monoclinic, and has four coppers and four oxygens in each unit cell, as shown in Figure 23.18. Density functional calculations of Ching et al. (1989)

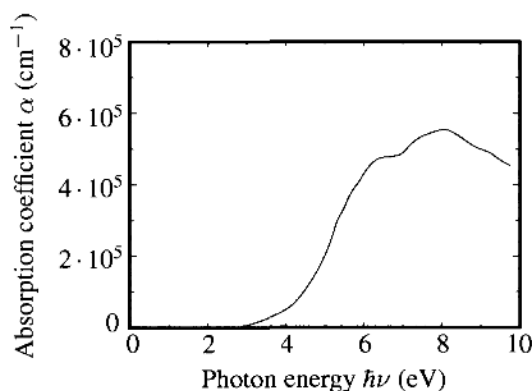


Figure 23.17. Optical absorption of CoO. The first traces of absorption appear at 3 eV, but these are attributed either to impurities or to excitonic effects. The steep linear rise, which extrapolates back to zero absorption at 4 eV, is thought to be more significant, and therefore the optical gap of CoO is conventionally said to be 4 eV. [Source: Powell and Spicer (1970), p. 2188.]

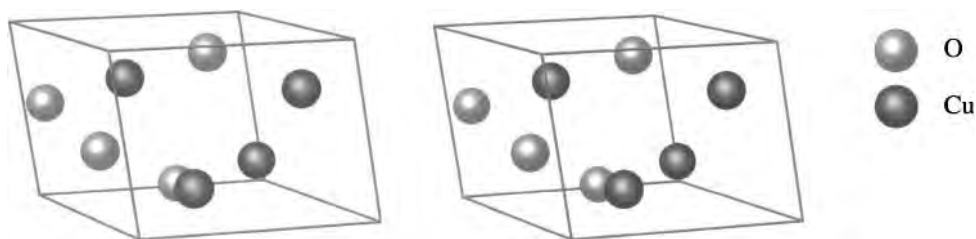


Figure 23.18. The structure of CuO, as determined by Åsbrink and Norrby (1970). The monoclinic unit cell contains 4 oxygens and 4 coppers. Stereo pair.

unambiguously predict CuO to be metallic, but in fact it is a semiconductor with a gap of 1.4 eV.

A great deal of research has been devoted to the transition metal oxides without resolving the essential difficulty. The general belief is that band theory fails because the d electrons are rather closely localized on the nickel atoms, and the density functional approximation underestimates the consequences of Coulomb repulsion between them. Copper oxide is on the insulating side of the metal–insulator transitions discussed in Section 18.3. The qualitative failure of single-electron band theory does not mean that there is no means to predict the results of experiments in CuO. A wide variety of experimental results can be described through the use of simple models to be described below. The use of these simple models is, however, predicated upon the knowledge that CuO is an insulator. They cannot predict that it belongs to the insulating class, nor explain why it does so.

The models used to explain CuO are *local*, which means essentially that they view the solid as a large molecule, where most of the physics can be understood by analyzing a tiny cluster of atoms, and the more remote atoms provide tiny additional perturbations. The starting points of these calculations are the energy levels of isolated copper and oxygen atoms, and they proceed by then considering what happens when the atoms are brought together in pairs. There is no trace of the wide range of propagating states, indexed by $\vec{k}$, that should characterize a metal.

These observations partially provide an explanation for why such a large number of core-level photoemission studies has been devoted to the transition metal oxides. A probe with a highly localized character is devoted to solids where the excitations have a similarly localized character. X-rays directed at copper are able to eject electrons from the $2p$ core state, which because of the spin–orbit interaction splits into $2p_{1/2}$ and $2p_{3/2}$ levels. In pure copper, the binding energy of $2p_{3/2}$ is $\mathcal{E}_{\text{core}} = 923.3$ eV. In CuO, not one but two peaks are found near this energy, but neither of the peaks has quite the expected energy. The differences in energies between the core binding energy $\mathcal{E}_{\text{core}}$ and the observed peak locations are due to changes in state of the valence electrons, and the fact that extra peaks are observed is due to the existence of multiple metastable valence states.

What are these different valence states? The answer is provided by a phenomenological form of quantum mechanics in which one uses a small number of

matrix elements as free parameters to fit the experimental data, guided mainly by energy levels of isolated atoms. The phenomenology of Fermi liquid theory began with the observation that metals exist, and it supposes that the actual quantum states remain in one-to-one correspondence with quantum states of the free Fermi gas. The logic in the case of CuO is similar, except that now by observing the solid to be an insulator, one makes the supposition that the actual quantum states will be in one-to-one correspondence with states of $\text{Cu}^{\text{II}}\text{O}^{-\text{II}}$ molecules.

Begin by considering quantum states of isolated atoms. Let $|d^{10}\rangle$ and $|d^9\rangle$ be the lowest-energy wave functions of a copper atom with 10 and 9 electrons in the d shell. In its ground state, copper has ten $3d$ electrons and one $4s$ electron, but oxygen is always able to steal at least the $4s$ electron from the copper. Similarly, let $|\text{O}^{-\text{I}}\rangle$, and $|\text{O}^{-\text{II}}\rangle$ be oxygen wave functions where the oxygen has acquired one or two extra electrons. A point that particularly needs to be emphasized is that these wave functions are many-electron wave functions. In particular, they take fully into account the energetic penalties that Coulomb repulsion imposes every time an additional electron is added to the atom. In the single-electron picture, adding two electrons of opposite spin to a solid costs the same energy for each, while adding two electrons of opposite spin to an atom can never act in that way.

Supposing these wave functions to be known, pass now to the CuO solid, but focus down on just a single adjoining copper and oxygen pair. Guess that the quantum states of the solid relevant for core-level photoemission are in one-to-one correspondence with quantum states of this molecule, and that the quantum states of the molecule can be constructed from linear combinations of the atomic wave functions. Considering only the wave functions where the total valence charge shared between copper and oxygen is conserved at 11 electrons, there are two states, $|d^9\text{O}^{-\text{II}}\rangle$ and $|d^{10}\text{O}^{-\text{I}}\rangle$. Let $\hat{\mathcal{H}}$ be the Hamiltonian for the cupric oxide solid, and take the expectation values of the Hamiltonian in these states to be

$$\langle d^9\text{O}^{-\text{II}}|\hat{\mathcal{H}}|d^9\text{O}^{-\text{II}}\rangle \equiv 0 \quad \begin{array}{l} \text{Energies are always arbitrary up to an} \\ \text{overall constant, so set this one to zero,} \\ \text{corresponding to the ground-state} \\ \text{energy of copper oxide.} \end{array} \quad (23.61a)$$

$$\langle d^{10}\text{O}^{-\text{I}}|\hat{\mathcal{H}}|d^{10}\text{O}^{-\text{I}}\rangle \equiv \Delta. \quad \text{Charge transfer energy.} \quad (23.61b)$$

The matrix element in Eq. (23.61b) is the *charge transfer energy*, the energy cost involved in moving an electron from oxygen to copper. Denote the off-diagonal component of $\hat{\mathcal{H}}$ in these states by

$$\langle d^9\text{O}^{-\text{II}}|\hat{\mathcal{H}}|d^{10}\text{O}^{-\text{I}}\rangle = \langle d^{10}\text{O}^{-\text{I}}|\hat{\mathcal{H}}|d^9\text{O}^{-\text{II}}\rangle \equiv T, \quad (23.62)$$

so that the low-energy excitations of CuO are given by the eigenvalues and eigenvectors of the matrix

$$\begin{pmatrix} 0 & T \\ T & \Delta \end{pmatrix}. \quad (23.63)$$

Problem 5 shows that the ground state is

$$|\Psi_{10}\rangle = \cos \theta_i |d^9\text{O}^{-\text{II}}\rangle - \sin \theta_i |d^{10}\text{O}^{-\text{I}}\rangle \quad (23.64a)$$

where

$$\tan 2\theta_i = \frac{2T}{\Delta}. \quad \text{When } T \text{ is much less than } \Delta, \text{ the ground state is almost pure } d^9, \text{ but as } T \text{ and } \Delta \text{ become comparable the two states mix.} \quad (23.64b)$$

Now imagine that an incoming photon knocks a $2p_{3/2}$ electron out of the core of the copper, and construct again the low-lying states of the valence electrons

$$\langle c^I d^9 O^{-II} | \hat{H} | c^I d^9 O^{-II} \rangle \equiv \mathcal{E}_{\text{core}} \quad \text{The energy needed to eject a core electron.} \quad (23.65a)$$

$$\langle c^I d^{10} O^{-I} | \hat{H} | c^I d^{10} O^{-I} \rangle \equiv \mathcal{E}_{\text{core}} + \Delta - U_{cd}. \quad \text{Charge transfer energy minus attraction to core hole.} \quad (23.65b)$$

The matrix elements of Eq. (23.65) are raised by the energy $\mathcal{E}_{\text{core}}$ needed to eject the core electron. In addition, each electron jumping onto the copper gains an energy U_{cd} through its Coulomb interaction with the positive charge in the core. Therefore U_{cd} displays in the simplest possible way the effect of electron correlation. In the single-electron picture, removing an electron from the core state could not change the relative energies of two other states, it would just shift them together.

Assuming off-diagonal terms of the Hamiltonian to be the same as before implies that valence states in the presence of the core hole are given by diagonalizing the matrix

$$\begin{pmatrix} \mathcal{E}_{\text{core}} & T \\ T & \mathcal{E}_{\text{core}} + \Delta - U_{cd} \end{pmatrix}. \quad (23.66)$$

Denote the eigenstates by

$$|\Psi_{f0}\rangle = \cos \theta_f |c^I d^9 O^{-II}\rangle - \sin \theta_f |c^I d^{10} O^{-I}\rangle \quad (23.67a)$$

$$|\Psi_{f1}\rangle = \sin \theta_f |c^I d^9 O^{-II}\rangle + \cos \theta_f |c^I d^{10} O^{-I}\rangle, \quad (23.67b)$$

where the label f indicates final states of the valence electrons and

$$\tan 2\theta_f = \frac{2T}{\Delta - U_{cd}}. \quad (23.67c)$$

To make use of the eigenvectors obtained from these matrices, consider the calculation of transition rates for the core-level photoemission experiment. The rate at which the process occurs is given by Eq. (20.72). The initial state of the system is given by the lowest-energy eigenstate of (23.63), and the final states are given by the eigenstates of (23.66). The matrix elements needing to be evaluated are

$$\langle c^0 | \hat{P} | c^I \rangle \langle \Psi_{i0} | \Psi_{f0,1} \rangle \quad \text{Because it is the core electron being ejected, the momentum operator acts upon the core electron, not on the valence electrons. The first matrix element places } \hat{P} \text{ in between core states with and without the ejected electron.} \quad (23.68)$$

The energy difference between the two eigenstates in Eq. (23.67) is

$$\Delta \mathcal{E} = \sqrt{(\Delta - U_{cd})^2 + 4T^2}, \quad (23.69)$$

and this energy should correspond to the peak splitting of CuO in Figure 23.19. From Eq. (23.68) one can estimate that the ratio of the heights of the two peaks in

Figure 23.19 should be (Problem 5)

$$\frac{|\langle \Psi_{i0} | \Psi_{f1} \rangle|^2}{|\langle \Psi_{i0} | \Psi_{f0} \rangle|^2} = \tan^2(\theta_i - \theta_f). \quad (23.70)$$

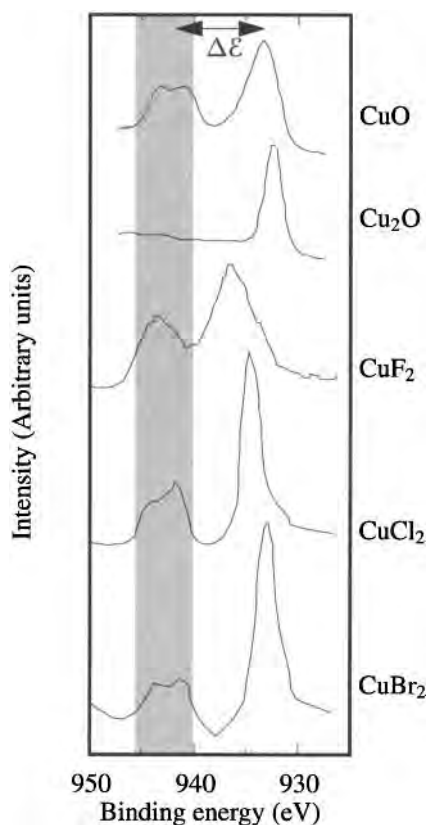


Figure 23.19. Core-level photoemission from CuO, Cu₂O, and copper dihalides, in the vicinity of the $2p_{3/2}$ state. As shown by the gray bar, the satellite line stays fixed for all compounds except Cu₂O, where it is absent, while the main line jumps about. [Source: Ghijsen et al. (1988), p. 11 324 and van der Laan et al. (1981), p. 4371.]

Completing the description of CuO consists in determining the parameters T , Δ , and U_{cd} that specify its low-energy excitations. This task cannot be accomplished through experiments on CuO alone. Core-level photoemission in the vicinity of $2p_{3/2}$ appears in the upper panel of Figure 23.19. Two peaks are visible. But how are they to be identified? For example, how can one tell whether $\langle c^{\uparrow}d^{10}\text{O}^{-\text{I}} |$ (d^{10} for short) or $\langle c^{\uparrow}d^9\text{O}^{-\text{II}} |$ (d^9 for short) lies lower in energy? These questions can be answered by comparing core-level photoemission of CuO with core-level spectra of other copper compounds such as Cu₂O and the copper dihalides CuF₂, CuCl₂, and CuBr₂.

Begin by comparing the photoemission intensity of CuO and Cu₂O. The essential difference between these two compounds is that in Cu₂O coppers can satisfy oxygen's longing for two electrons by donating one each from the $4s$ states, leaving the $3d^{10}$ bands intact. Only one peak appears in the photoemission spectrum of Cu₂O, and it must be identified with a predominantly $3d^{10}$ final state of copper. The left-hand peak in all the other compounds must then be identified with a predominantly $3d^9$ final state. In the processes that create the left-hand peak, an

X-ray photon hits a $2p_{3/2}$ core electron of a copper atom whose outer shell is in state $3d^9$, and the outer shell remains in state $3d^9$ as the electron is ejected. This process should be insensitive to the atoms surrounding the copper, and accordingly the energy of this peak changes little as one moves from compound to compound. For the right-hand peak, ejection of the core electron is accompanied by capture of an electron from a neighboring oxygen to form $3d^{10}$. This second process depends upon the interaction with neighbors, and therefore should vary in energy from compound to compound. The rightmost peak in Figure 23.19 does in fact move about in energy more noticeably than the satellite.

Having determined the nature of the peaks in Figure 23.19, there still remains the problem that the three parameters Δ , T , and U_{cd} cannot be determined uniquely from the two equations (23.69) and (23.70). Problem 6 shows how to combine information from the different compounds to provide a plausible solution. Assuming that T and U_{cd} are approximately the same for all compounds while Δ varies, one can determine that $U_{cd} \approx 9$ eV, $T \approx 2.5$ eV, and that for CuO, $\Delta \approx 1.5$ eV.

Problems

1. Qualitative optical properties:

- What is the frequency of electromagnetic radiation at which aluminum should become transparent, according to the Drude theory?
- What is the frequency of electromagnetic radiation at which very pure silicon, at 300 K, should become transparent according to the Drude theory?
- Pure silicon has a silvery mirror-like surface. Provide a qualitative explanation for why it is not transparent.
- Draw a sketch of the absorption coefficient of silicon from $\omega = 0$ up to the visible.

2. Optical absorption of alkali metals: The aim of this problem is to verify Eq. (23.48). The calculation is straightforward, except that expressions such as Eq. (20.70) have a very condensed notation, and it can be confusing to untangle them.

- Verify that Eqs. (23.45) and (23.46) follow from Eq. (8.14) when one focuses upon a single Bragg vector $\vec{K}$.
- In the present case, one really needs to consider the 12 equivalent vectors $\langle 110 \rangle$. How should Eqs. (23.45) and (23.46) be generalized?
- Show that Eq. (23.47) follows from Eq. (20.70).
- Carry out the integrals needed to verify Eqs. (23.50) and (23.51).
- Find numerical values for ω^{low} and ω^{high} in the case of sodium, in units of Hz, and also evaluate the joint density of states in units of $1/[\text{eV cm}^3]$.

- (f) Referring to Figure 23.7, estimate the pseudopotential energy $|U_{\vec{K}}|$ for $\vec{K} = (110)$.

3. **Helicon waves:** Plasma oscillations constitute the sloshing back and forth of a cloud of electrons. In the presence of a static magnetic field, the calculation becomes more complicated. Assume a magnetic field $\vec{B}_0$ is present in a conducting material, pointing along the $\hat{z}$ axis, and one passes an electromagnetic wave along the same axis. An analog of the purely classical calculation of the plasma frequency may be obtained as follows:

- (a) Begin with Maxwell's equations

$$\nabla \times \vec{E} = \frac{-1}{c} \frac{\partial \vec{B}}{\partial t} \quad (23.71)$$

$$\nabla \times \vec{B} = \frac{1}{c} \frac{\partial \vec{E}}{\partial t} + \frac{4\pi}{c} \vec{j}, \quad (23.72)$$

with

$$\vec{j}(\omega) = \sigma(\omega) \vec{E}(\omega). \quad (23.73)$$

Assume that the electric field propagates along the $\hat{z}$ axis as

$$\vec{E} = \vec{E}_0 e^{i\vec{k} \cdot \vec{r} - i\omega t} \quad (23.74)$$

with

$$\vec{k} = (0, 0, k). \quad (23.75)$$

Define

$$k_0 = \frac{\omega}{c}. \quad (23.76)$$

Show first that

$$\vec{k} \times \vec{k} \times \vec{E}_0 + k_0^2 \epsilon \vec{E}_0 = 0 \quad (23.77)$$

and find an expression for the dielectric tensor ϵ in terms of the conductivity tensor σ .

- (b) Assuming rotational symmetry around the $\hat{z}$ axis, so that x and y are equivalent, and assuming further that

$$\epsilon_{xz} = \epsilon_{yz} = 0 \quad (23.78)$$

because electrons move in circles in the x - y plane only, show that Eq. (23.77) becomes

$$\left(\frac{k^2}{k_0^2} - \epsilon_+ \right) \left(\frac{k^2}{k_0^2} - \epsilon_- \right) \epsilon_{zz} = 0 \quad (23.79)$$

and find ϵ_+ and ϵ_- in terms of ϵ_{xx} and ϵ_{xy} .

(c) Take

$$\vec{j} = -ne\vec{v} \quad (23.80)$$

and

$$m\dot{\vec{v}} = -e \left[\vec{E} + \frac{\vec{v}}{c} \times \vec{B}_0 \right] - \frac{m\vec{v}}{\tau}. \quad (23.81)$$

Recall that $\vec{B}_0$ is the static field along the $\hat{z}$ axis. Solve Eqs. (23.73), (23.80), and (23.81) for the dielectric tensor, and as a consequence find the frequency of the helicon modes. You should find along the way that

$$\epsilon_{xx} = 1 - \frac{\omega_p^2}{\omega\tilde{\omega}} \frac{1}{[1 - (\omega_c/\tilde{\omega})^2]}, \quad (23.82)$$

where

$$\omega_c = \frac{eB_0}{mc} \quad (23.83)$$

and

$$\tilde{\omega} = \omega + i/\tau. \quad (23.84)$$

- (d) Find the values of ω_c and ω_p for aluminum in $B_0 = 10^4 G$. For ω comparable to the cyclotron frequency, show that the first term on the right-hand side of Eq. (23.82) can be neglected.
- (e) Find the dispersion relation for helicon waves, and say whether they propagate when ω is less than or greater than the cyclotron frequency

4. Landau damping:

- (a) The susceptibility of an electron gas is given by Eq. (23.33). Argue that χ will acquire an imaginary part when the denominator of Eq. (23.33) can vanish, and find an expression for the smallest value of $\vec{q}$ for which it is possible.
- (b) Evaluate this expression for aluminum

5. Core-level photoemission I:

- (a) Verify Eqs. (23.64) and (23.67).
- (b) Verify Eqs. (23.69) and (23.70).

6. Core-level photoemission II:

- (a) For each of the copper dihalides in Figure 23.19, take T to be a free parameter, and using Eqs. (23.69) and (23.70), plot U_{cd} versus T . Make use of the energy gap $\Delta\mathcal{E}$ and the ratio of peak heights in order to determine Δ as a function of T . Show that the curves intersect for $U_{cd} \approx 9$ eV; therefore, give U_{cd} this value.
- (b) Find Δ for CuO.
- (c) The optical band gap of CuO can now be estimated based upon a property of the matrix (23.63). Decide what calculation should correspond to this physical quantity, and find the optical band gap.

References

- F. Abelès, ed. (1972), *Optical Properties of Solids*, North-Holland, Amsterdam.
- S. Åsbrink and L.-J. Norrby (1970), A refinement of the crystal structure of copper (II) oxide with a discussion of some exceptional e.s.d.'s, *Acta Crystallographica B*, **26**, 8–15.
- L. Brillouin (1922), Scattering of light and X-rays by a transparent homogeneous body: Influence of thermal agitation, *Annales de Physique*, **17**, 88–122. In French.
- J. R. Chelikowsky and M. L. Cohen (1976), Nonlocal pseudopotential calculations for the electronic structure of eleven diamond and zinc-blende semiconductors, *Physical Review B*, **14**, 556–582.
- T.-C. Chiang, J. A. Knapp, M. Aono, and D. E. Eastman (1980), Angle-resolved photoemission, valence-band dispersions $E(k)$, and electron and hole lifetimes for GaAs, *PRB*, **21**, 3513–3522.
- W. Y. Ching, Y.-N. Xu, and K. W. Wong (1989), Ground-state and optical properties of Cu_2O and CuO crystals, *Physical Review B*, **40**, 7684–7695.
- B. Dörner, E. Burkel, T. Illini, and J. Peisl (1987), First measurement of a phonon dispersion curve by inelastic X-ray scattering, *Zeitschrift für Physik B*, **69**, 179–183.
- J. Ghijsen, H. Tjeng, J. van Elp, H. Eskes, J. Westerink, G. A. Sawatzky, and M. T. Czyzyk (1988), Electronic structure of Cu_2 and CuO , *Physical Review B*, **38**, 11 322–11 330.
- M. Glicksman (1971), Plasmas in solids, *Solid State Physics: Advances in Research and Applications*, **26**, 275–427.
- V. Heine (1980), Electronic structure from the point of view of the local atomic environment, *Solid State Physics: Advances in Research and Applications*, **35**, 1–127.
- C. H. Henry and J. J. Hopfield (1965), Raman scattering by polaritons, *Physical Review Letters*, **15**, 964–966.
- F. J. Himpsel and N. V. Smith (1985), Photoelectron spectroscopy, *Physics Today*, **38**(12), 60–66.
- J. Hölzl and F. K. Schulte (1979), Work function of metals, in *Solid Surface Physics*, vol. 85 of *Springer Tracts in Modern Physics*, Springer-Verlag, Berlin.
- S. Hüfner (1994), Electronic structure of NiO and related 3d-transition-metal compounds, *Advances in Physics*, **43**, 183–356.
- E. D. Isaacs and P. M. Platzman (1996), Inelastic X-ray scattering in condensed matter systems, *Physics Today*, **49**(2), 40–45.
- E. Jensen, R. A. Bartynski, T. Gustafsson, E. W. Plummer, M. Y. Chou, M. L. Cohen, and G. B. Hoflund (1984), Angle-resolved photoemission study of the electronic structure of beryllium: Bulk band dispersions and many-electron effects, *Physical Review B*, **30**, 5500–5507.
- C. Kunz (1962), On the angular dependence of characteristic energy loss in Al, Si, Ag, *Zeitschrift für Physik*, **167**, 53–71. In German.
- H. Landolt and R. Börnstein (New Series), *Numerical Data and Functional Relationships in Science and Technology*, New Series, Group III, Springer-Verlag, Berlin.
- W. Lang (1948), Velocity loss of medium speed electrons during passage through thin metal foils, *Optik (Stuttgart)*, **3**, 233–246. In German.
- S. G. Louie (1992), Quasiparticle excitations and photoemission, in *Angle-Resolved Photoemission: Theory and Current Practice*, S. D. Kevan, ed., vol. 74 of *Studies in Surface Science and Catalysis*, pp. 33–98, Elsevier, Amsterdam.
- S. T. Manson and A. F. Starace (1982), Photoelectron angular distributions: Energy dependence for s subshells, *Reviews of Modern Physics*, **54**, 389–405.
- G. Margaritondo (1988), 100 years of photoemission, *Physics Today*, **41**(4), 66–72.
- E. G. McRae (1979), Electronic surface resonances of crystals, *Reviews of Modern Physics*, **51**, 541–568.
- N. F. Mott (1949), The basis of the electron theory of metals with special reference to the transition metals, *Proceedings of the Physical Society (London)*, **62 A**, 416–422.
- K. Ohtaka and Y. Tanabe (1990), Theory of the soft-X-ray edge problem in simple metals: Historical

- survey and recent developments, *Reviews of Modern Physics*, **62**, 929–991.
- K. C. Pandey and J. C. Phillips (1974), Nonlocal pseudopotentials for Ge and GaAs, *Physical Review B*, **9**, 1552–1559.
- F. Patthey, J. M. Imer, W. D. Schneider, H. Beck, Y. Baer, and B. Delly (1990), High-resolution photoemission study of the low-energy excitations in 4f-electron systems, *Physical Review B*, **42**, 8864–8881.
- A. B. Pippard (1957), An experimental determination of the Fermi surface in copper, *Philosophical Transactions of the Royal Society of London*, **A250**, 325–357.
- A. B. Pippard (1988), *Magnetoresistance in Metals*, Cambridge University Press, Cambridge.
- P. M. Platzman and P. A. Wolff (1973), *Waves and Interactions in Solid State Plasmas*, Academic Press, New York.
- R. J. Powell and W. E. Spicer (1970), Optical properties of NiO and CoO, *Physical Review B*, **2**, 2182–2193.
- C. V. Raman (1928), A change of wavelength in light scattering, *Nature*, **121**, 619–620.
- D. H. Rich, T. Miller, G. E. Franklin, and T.-C. Chiang (1989), Sb-induced bulk band transitions in Si(111) and Si(001) observed in synchrotron photoemission studies, *Physical Review B*, **39**, 1438–1441.
- G. Ruthemann (1948), Discrete energy loss of medium velocity electrons during passage through thin foils, *Annals of Physics*, **2**, 114–134. In German.
- J. R. Sandercock (1972), Brillouin-scattering measurements on silicon and germanium, *Physical Review Letters*, **28**, 237–240.
- S. E. Schnatterly (1979), Inelastic electron scattering spectroscopy, *Solid State Physics: Advances in Research and Applications*, **34**, 275–358.
- N. V. Smith (1970), Photoemission studies of the alkali metals. ii: Rubidium and cesium, *Physical Review B*, **3**, 3662–3670.
- D. Straub, L. Ley, and F. J. Himpsel (1986), Inverse-photoemission study of unoccupied electronic states in Ge and Si: Bulk energy bands, *Physical Review B*, **33**, 2607–2614.
- M. L. Thèye (1968), Ph.D. thesis, University of Paris VI. In French.
- G. van der Laan, C. Westra, C. Haas, and G. A. Sawatzky (1981), Satellite structure in photoelectron and Auger spectra of copper dihalides, *Physical Review B*, **23**, 4369–4380.
- A. L. Wachs, T. Miller, T. C. Hsieh, A. P. Shapiro, and T.-C. Chiang (1985), Angle-resolved photoemission studies of Ge(111)-(2×8), Ge(111)-(1×1)H, Si(111)-(7×7), and Si(100)-(2×1), *Physical Review B*, **32**, 2326–2333.
- G. P. Williams, F. Cerrina, G. J. Lapeyre, J. R. Anderson, R. J. Smith, and J. Hermanson (1986), Experimental study of the band structure of GaP, GaAs, GaSb, InP, InAs, and InSb, *Physical Review B*, **34**, 5548–5557.